R127/model_only_seed2 | Single VAL-best full18 results

R83 model-only source; MVMOE removed; BQ/LEHD retained; same VAL lock; FP64 argmin.

epoch 26 | update 37440 | seed2 | fresh initialization

Data: selection_data_new90k_no_mvmoe_seed2_20261112

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	81.1296	76.3580	11.417495	11.427978	11.398095	-0.1881	+0.2082	0.2252
TEST ALL	81.2704	76.1457	11.374453	11.385299	11.355153	-0.1929	+0.2076	0.2228

Dataset	n	Fixed NSS Top1 (%)	R127 Top1 (%)	Fixed NSS cost	R127 cost
TSP TEST	4500	45.8889	46.6889	7.97040072	7.97185218
CVRP TEST	4500	46.7333	48.6444	18.00537547	18.01148229
TSPLIB	49	20.4082	26.5306	8.21790707	8.21187632
CVRPLIB	100	69.0000	58.0000	67.10265619	67.21106256

NSS: fixed historical logits reselected after deleting MVMOE; NOT a matched reduced-pool training control.

Only existing aligned4500-row TEST / original LIB predictions are reused; unavailable entries are MISSING.

SBS: reduced-pool full81000 TRAIN only. Oracle: minimum legal FP64 cost per instance.

ALL: equal weight per problem. Macro raw costs are descriptive, not cross-problem pooled costs.

TRAIN81000 / VAL4500 / TEST4500 per problem. Retained holdouts informed earlier R&D, not fully blind.

R83 model-only architecture reference: older data / original pool, NOT a matched training control.

Deletion changes Oracle and winner labels; a Top1 change alone does not establish an architectural cause.

R127/model_only_seed2 | VAL: ALL and18 tasks

Top1 higher is better; cost/regret lower is better. Same checkpoint for every task.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	81.1296	76.3580	11.417495	11.427978	11.398095	-0.1881	+0.2082	0.2252	per-problem TRAIN SBS
TSP	46.6222	33.1111	8.024783	8.045354	7.990533	-0.2557	+0.4286	0.4170	T2T
CVRP	46.5778	19.0000	18.160836	18.269377	18.065094	-0.5941	+0.5300	0.7050	RELD_CVRP
ATSP	77.2444	47.4889	1.638324	1.676905	1.632471	-2.3007	+0.3585	0.3554	UNICO_MatPOENet
OVRP	40.8889	35.8889	7.462182	7.469579	7.415913	-0.0990	+0.6239	0.6360	MoSES_RF
VRPB	93.8889	93.9333	9.354654	9.354876	9.339112	-0.0024	+0.1664	0.1776	reld_mt
VRPL	38.3778	33.9556	12.430305	12.437959	12.384843	-0.0615	+0.3671	0.3748	MoSES_RF
VRPTW	98.2667	98.2000	18.819187	18.819431	18.812672	-0.0013	+0.0346	0.0490	reld_mt
OVRPTW	98.6889	98.6889	10.143730	10.143730	10.140192	+0.0000	+0.0349	0.0447	reld_mt
OVRPB	93.8222	93.8889	6.358712	6.357254	6.348484	+0.0229	+0.1611	0.1758	reld_mt
OVRPL	41.7778	36.1111	7.438632	7.445709	7.390783	-0.0951	+0.6474	0.6657	MoSES_RF
VRPBL	93.2222	93.2222	9.371433	9.371523	9.355289	-0.0010	+0.1726	0.1882	reld_mt
VRPBTW	99.8444	99.8444	19.950430	19.950430	19.950001	+0.0000	+0.0021	0.0030	reld_mt
VRPLTW	97.7111	97.7111	18.765871	18.765871	18.757204	+0.0000	+0.0462	0.0606	reld_mt
OVRPBL	94.7333	94.7111	6.370106	6.370093	6.361280	+0.0002	+0.1387	0.1509	reld_mt
OVRPBTW	99.9778	99.9778	10.661098	10.661098	10.661066	+0.0000	+0.0003	0.0005	reld_mt
OVRPLTW	98.8444	98.8667	10.157627	10.157423	10.154330	+0.0020	+0.0325	0.0463	reld_mt
VRPBLTW	99.8444	99.8444	19.797044	19.797044	19.796493	+0.0000	+0.0028	0.0028	reld_mt
OVRPBLTW	100.0000	100.0000	10.609955	10.609955	10.609955	+0.0000	+0.0000	0.0000	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; NOT a ratio of means.

Native FP64 argmin after candidate deletion; never FP32 cost argmin.

B tasks K=2: Top2/Top3 both cover all candidates (min(k,K)); 100% is not extra ranking evidence.

R127/model_only_seed2 | TEST: ALL and18 tasks

Top1 higher is better; cost/regret lower is better. Same checkpoint for every task.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	81.2704	76.1457	11.374453	11.385299	11.355153	-0.1929	+0.2076	0.2228	per-problem TRAIN SBS
TSP	46.6889	34.5333	7.971852	7.989621	7.937593	-0.2224	+0.4316	0.4228	T2T
CVRP	48.6444	18.4000	18.011482	18.124983	17.919470	-0.6262	+0.5135	0.6685	RELD_CVRP
ATSP	77.0889	47.2444	1.635168	1.674207	1.629441	-2.3318	+0.3514	0.3494	UNICO_MatPOENet
OVRP	40.9111	33.8444	7.446722	7.455750	7.400732	-0.1211	+0.6214	0.6326	MoSES_RF
VRPB	93.0000	93.0444	9.351756	9.351810	9.335507	-0.0006	+0.1741	0.1869	reld_mt
VRPL	38.5778	33.2222	12.348752	12.356138	12.302506	-0.0598	+0.3759	0.3844	MoSES_RF
VRPTW	98.4000	98.4000	18.690538	18.690545	18.683013	-0.0000	+0.0403	0.0524	reld_mt
OVRPTW	98.8889	98.9111	10.093088	10.093063	10.090708	+0.0003	+0.0236	0.0287	reld_mt
OVRPB	93.7778	93.6667	6.371178	6.370826	6.360658	+0.0055	+0.1654	0.1772	reld_mt
OVRPL	42.5111	34.9333	7.464218	7.472542	7.418418	-0.1114	+0.6174	0.6271	MoSES_RF
VRPBL	93.6444	93.6000	9.395525	9.396379	9.378949	-0.0091	+0.1767	0.1885	reld_mt
VRPBTW	99.8000	99.8000	19.665725	19.665725	19.664979	+0.0000	+0.0038	0.0052	reld_mt
VRPLTW	97.8889	97.9333	18.803228	18.803136	18.793442	+0.0005	+0.0521	0.0720	reld_mt
OVRPBL	94.4222	94.4889	6.376016	6.375741	6.366192	+0.0043	+0.1543	0.1706	reld_mt
OVRPBTW	100.0000	100.0000	10.659591	10.659591	10.659591	+0.0000	+0.0000	0.0000	reld_mt
OVRPLTW	98.8222	98.8000	10.086093	10.086112	10.082902	-0.0002	+0.0316	0.0411	reld_mt
VRPBLTW	99.8000	99.8000	19.707068	19.707068	19.706504	+0.0000	+0.0029	0.0036	reld_mt
OVRPBLTW	100.0000	100.0000	10.662149	10.662149	10.662149	+0.0000	+0.0000	0.0000	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; NOT a ratio of means.

Native FP64 argmin after candidate deletion; never FP32 cost argmin.

B tasks K=2: Top2/Top3 both cover all candidates (min(k,K)); 100% is not extra ranking evidence.

R127/model_only_seed2 | Native LIB and provenance

Original LIB inputs/costs; MVMOE removed; winner recomputed in FP64; no new solver answers.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	26.5306	10.2041	8.211876	8.240435	8.142167	-0.3466	+0.8562	0.9066
CVRPLIB	58.0000	46.0000	67.211063	67.245364	66.945727	-0.0510	+0.3963	0.5499

SHA256: 389dcc8b1817e6c9852dc7eee052c4caec5da49eedda019e34269dae24b3a641

Selection: strict minimum full18 VAL macro_vs_sbs_pct; no-MVMOE full TRAIN SBS; first wins ties

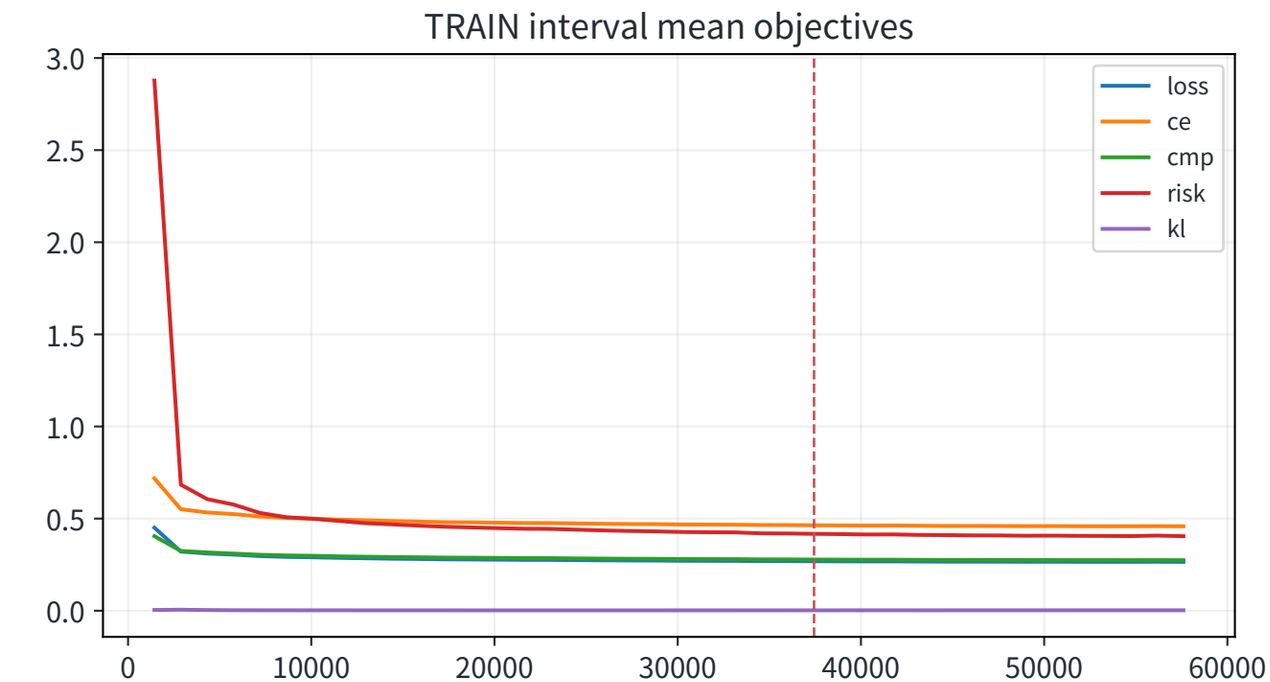
Factory: code.V4.r127_experiment.load_checkpoint

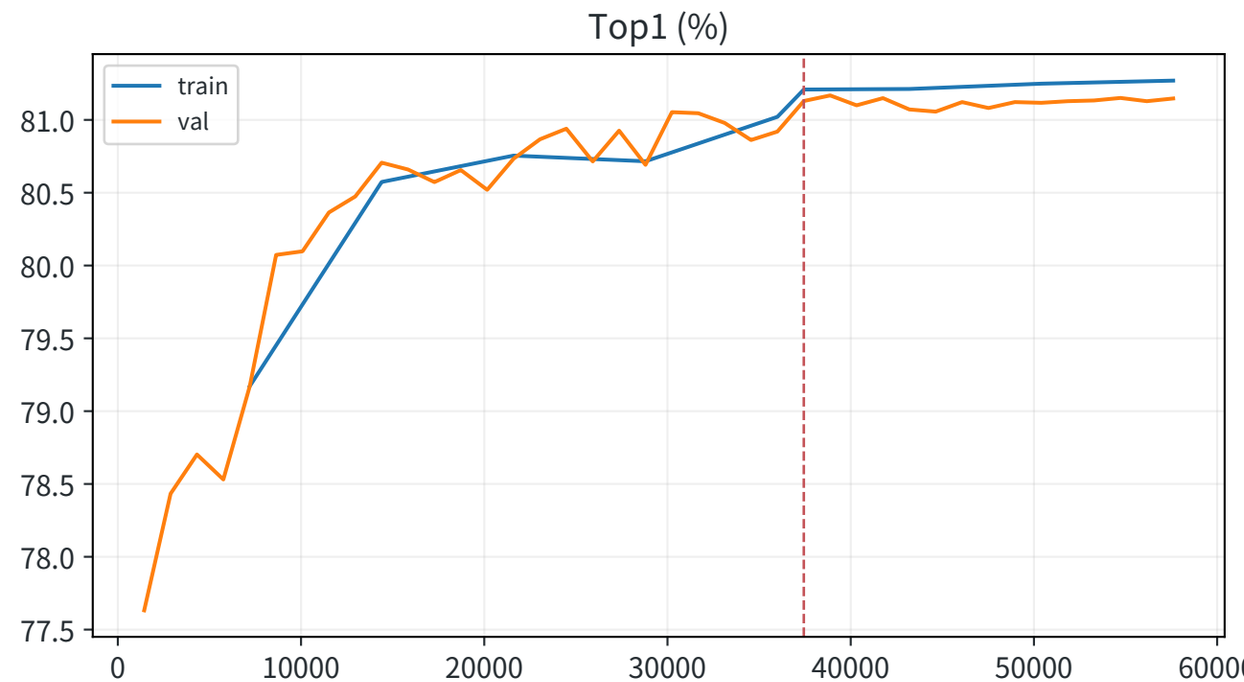
Inference: original utility score differences and maximin; no MVMOE; ascending-global-ID tie break

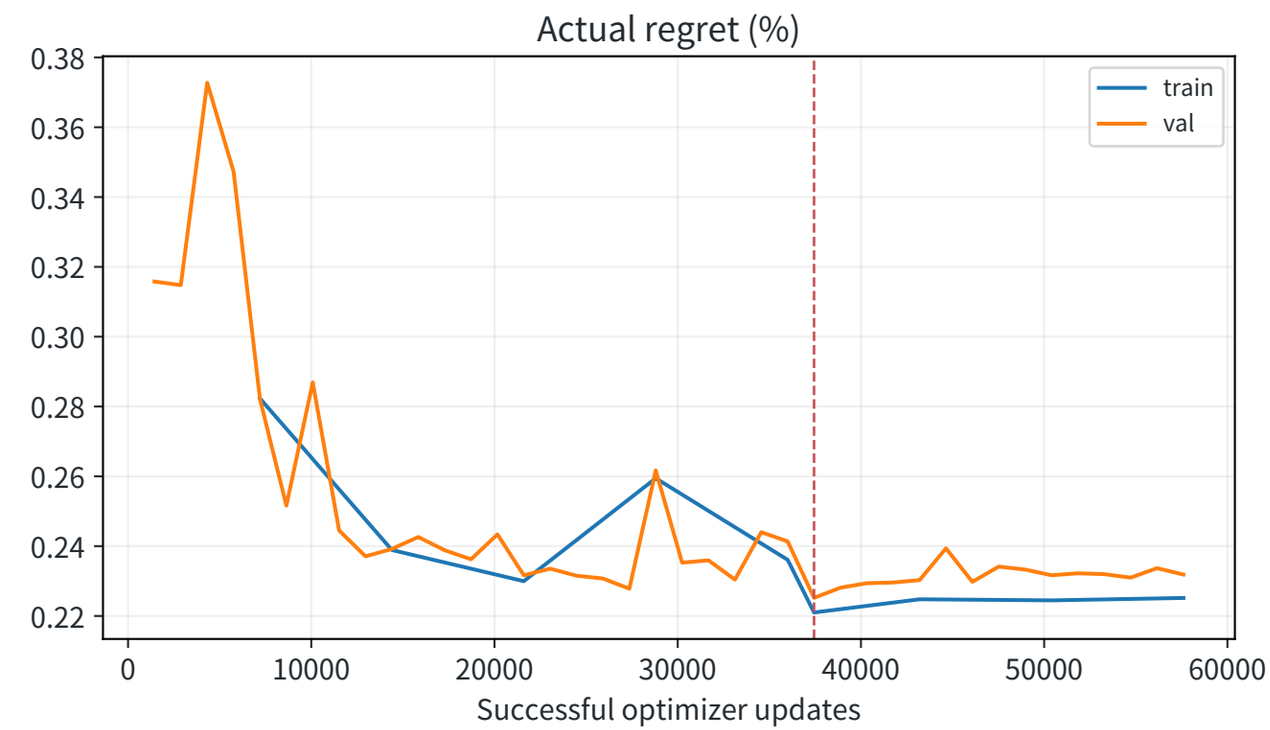
TSP8 / CVRP9 / ATSP5 / non-B5 / B2; BQ and LEHD retained.

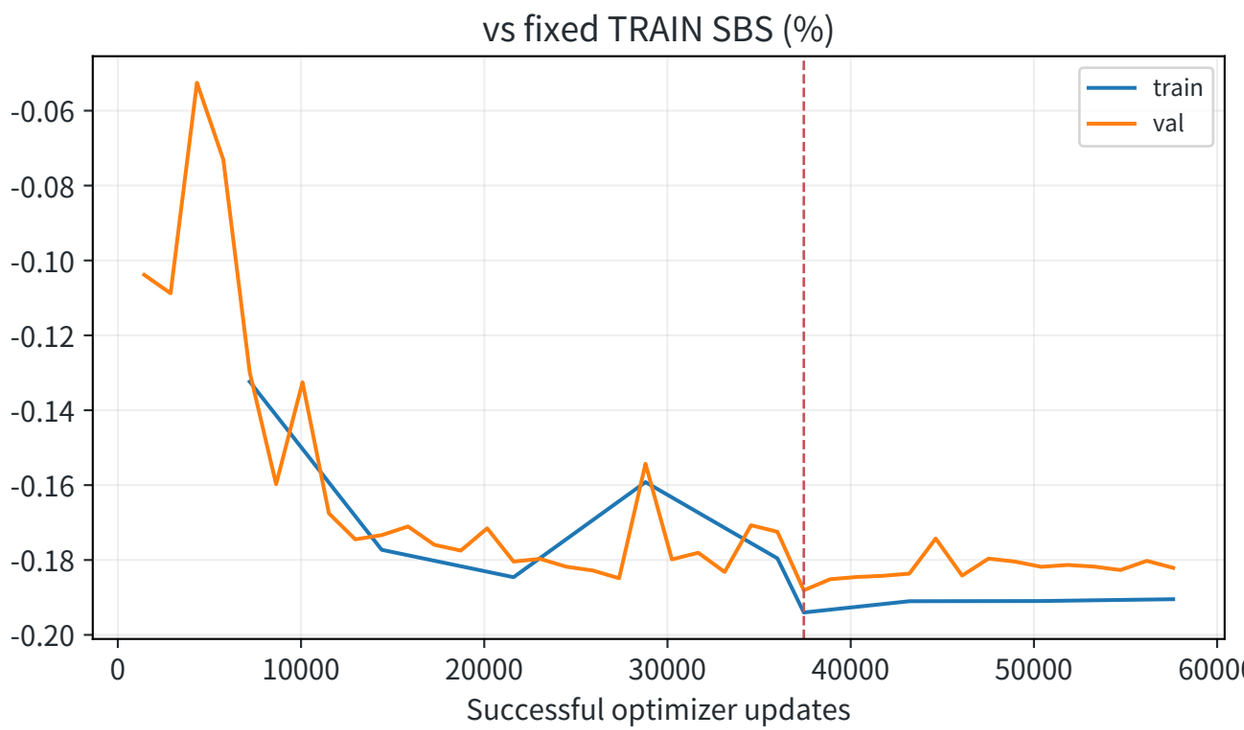
encoder+decision active; inference/config absent; GLOP ID-only. TRAIN buffers; no LIB refit.

R127/model_only_seed2 | one locked VAL checkpoint







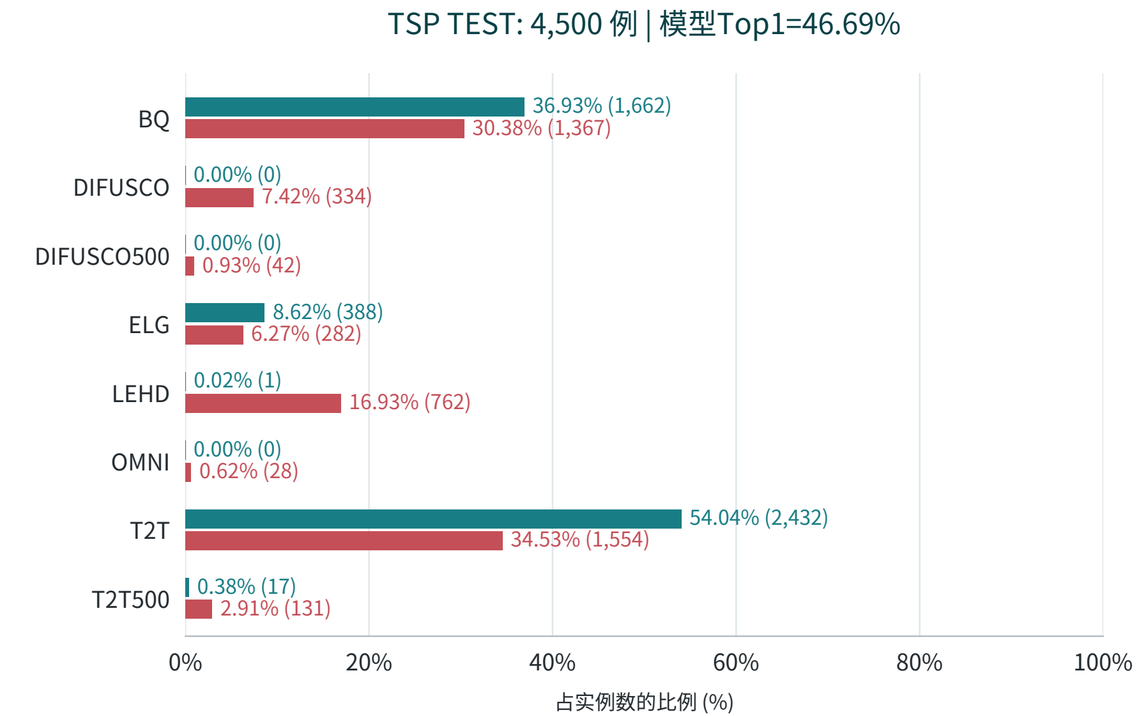


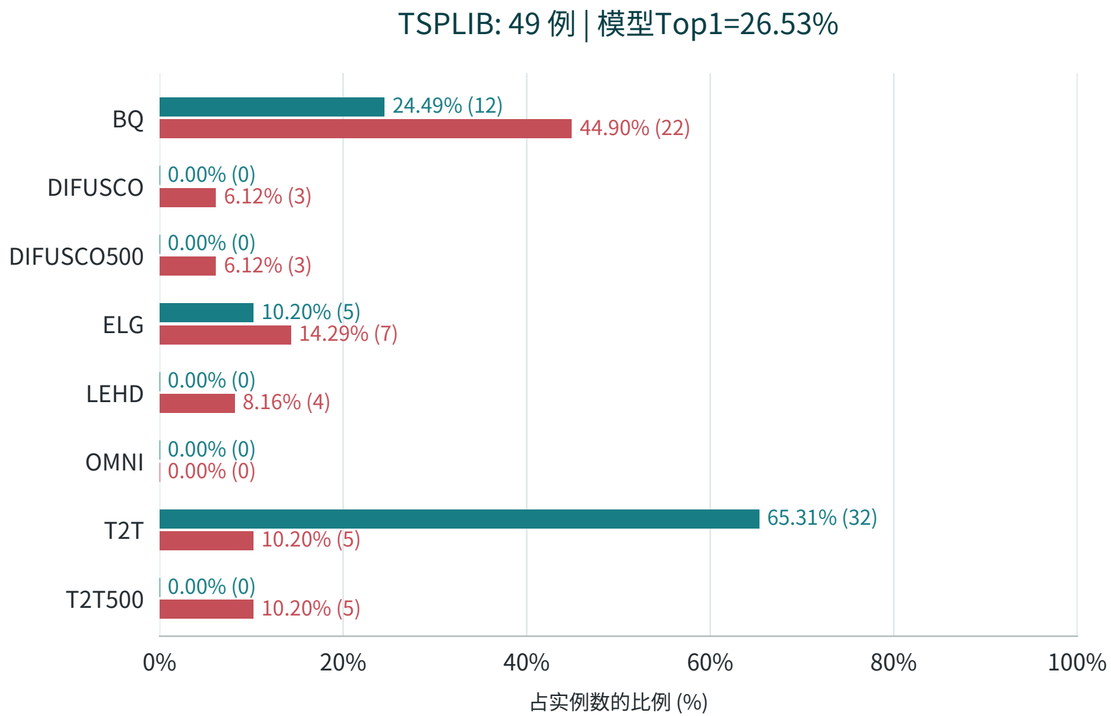
R127/model_only_seed2 | TSP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 389dcc8b1817e6c9

新 TEST 数据版本: selection_data_new90k_no_mvmoe_seed2_20261112

LIB: data/TSPLIB | FP64 argmin after MVMOE deletion





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

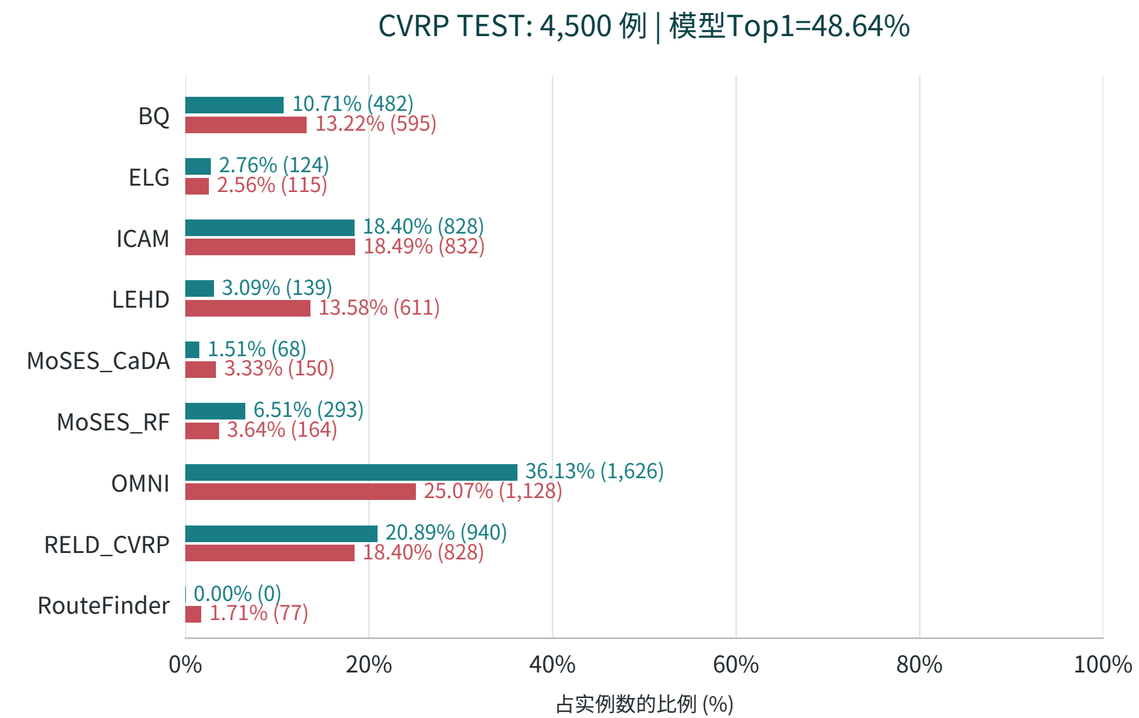
R127: only MVMOE deleted; BQ/LEHD and originally legal MoSES_RF retained; same VAL lock.

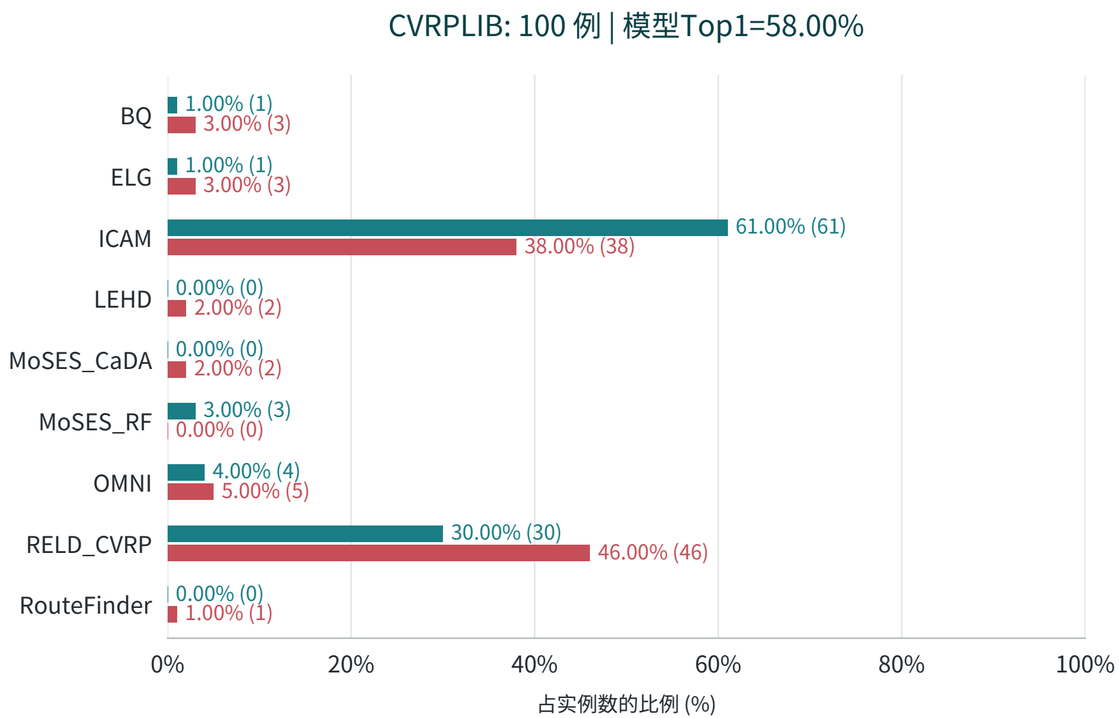
R127/model_only_seed2 | CVRP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 389dcc8b1817e6c9

新 TEST 数据版本: selection_data_new90k_no_mvmoe_seed2_20261112

LIB: data/CVRPLIB | FP64 argmin after MVMOE deletion





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

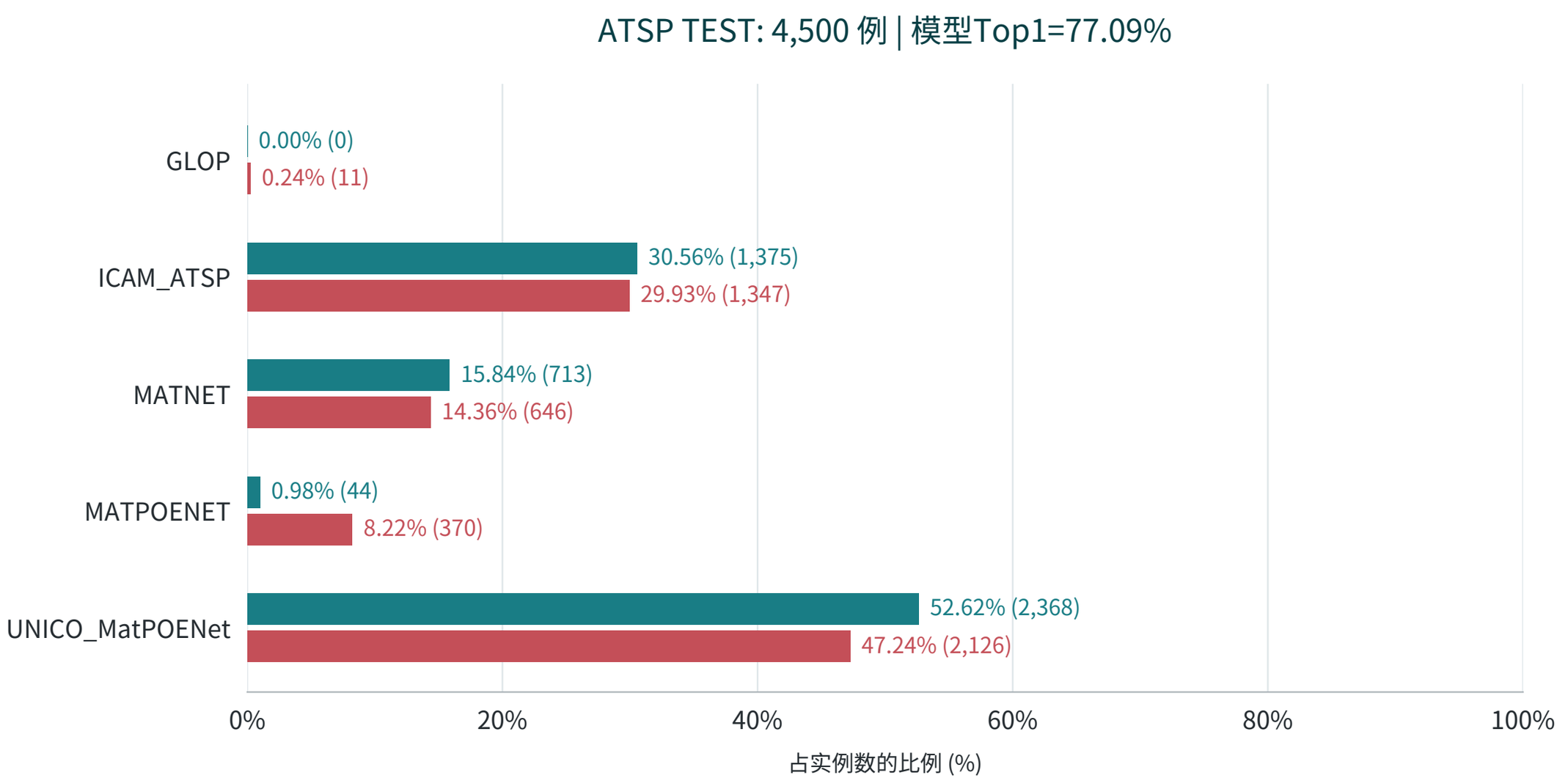
R127: only MVMOE deleted; BQ/LEHD and originally legal MoSES_RF retained; same VAL lock.

R127/model_only_seed2 | ATSP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 389dcc8b1817e6c9

新 TEST 数据版本: selection_data_new90k_no_mvmoe_seed2_20261112

标签: 原生 FP64 argmin after MVMOE deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

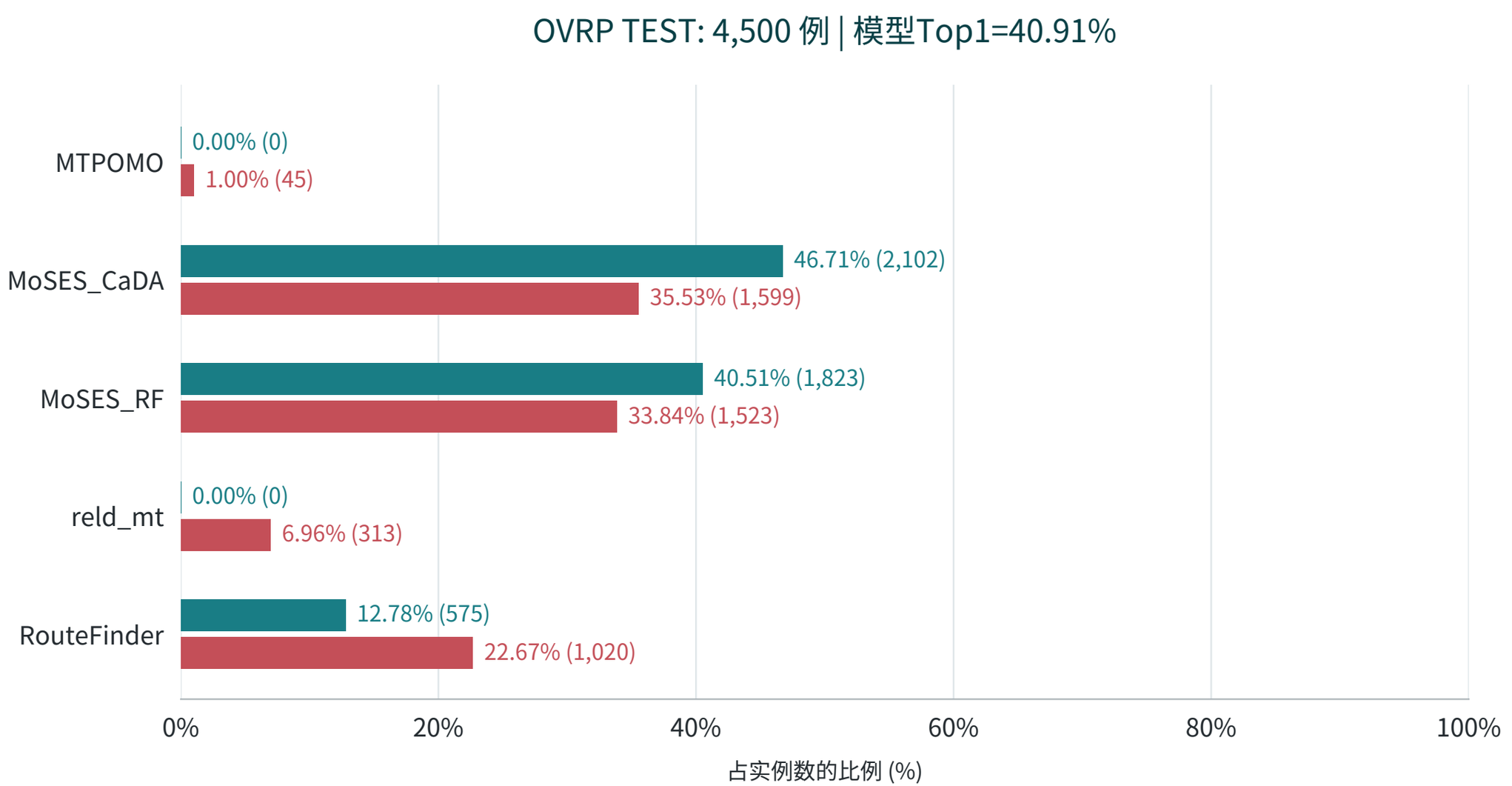
R127 ATSP: GLOP has no source roles; original trainable solver ID only.

R127/model_only_seed2 | OVRP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 389dcc8b1817e6c9

新 TEST 数据版本: selection_data_new90k_no_mvmoe_seed2_20261112

标签: 原生 FP64 argmin after MVMOE deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

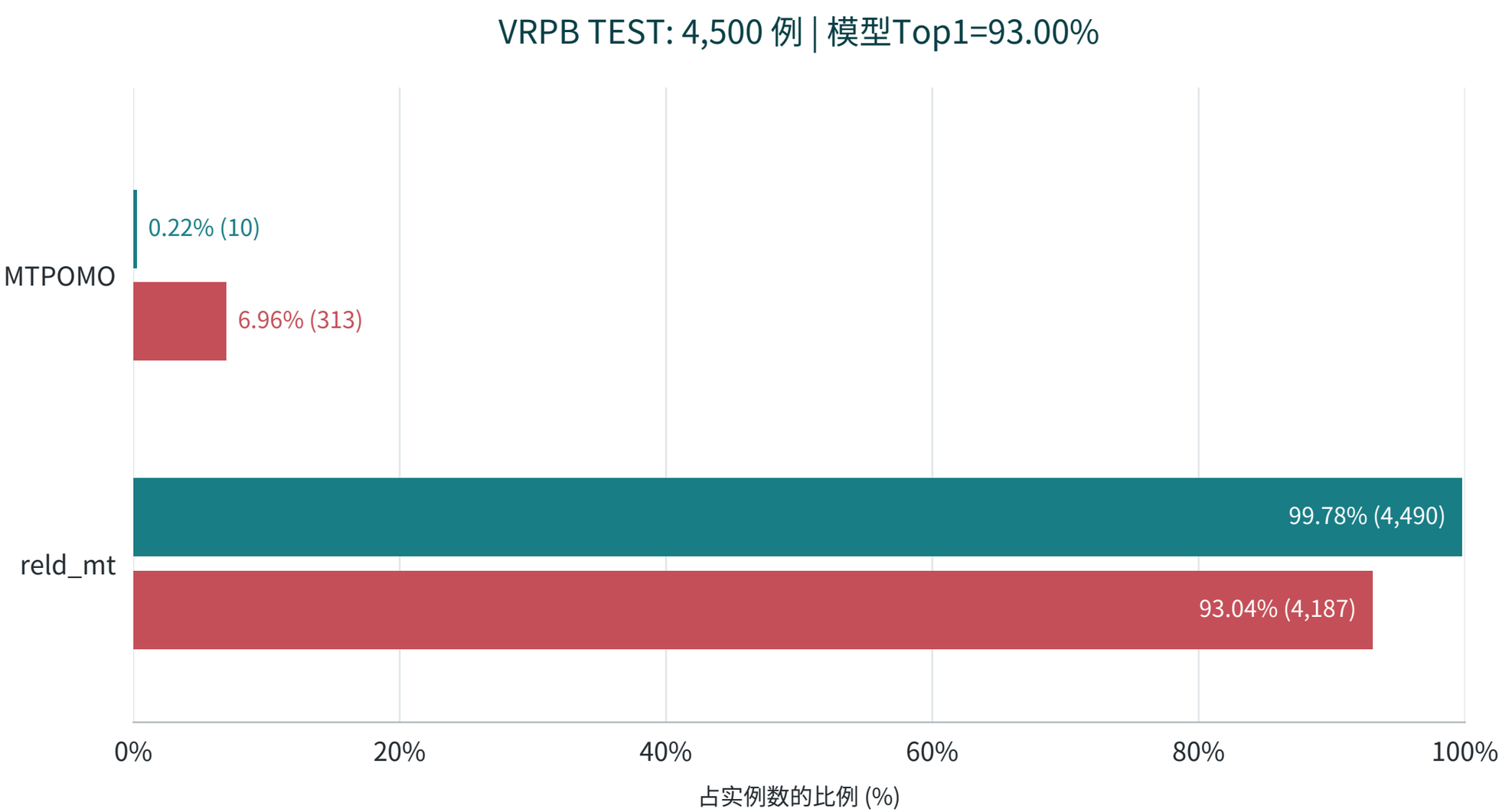
R127: only MVMOE deleted; BQ/LEHD and originally legal MoSES_RF retained; same VAL lock.

R127/model_only_seed2 | VRPB | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 389dcc8b1817e6c9

新 TEST 数据版本: selection_data_new90k_no_mvmoe_seed2_20261112

标签: 原生 FP64 argmin after MVMOE deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

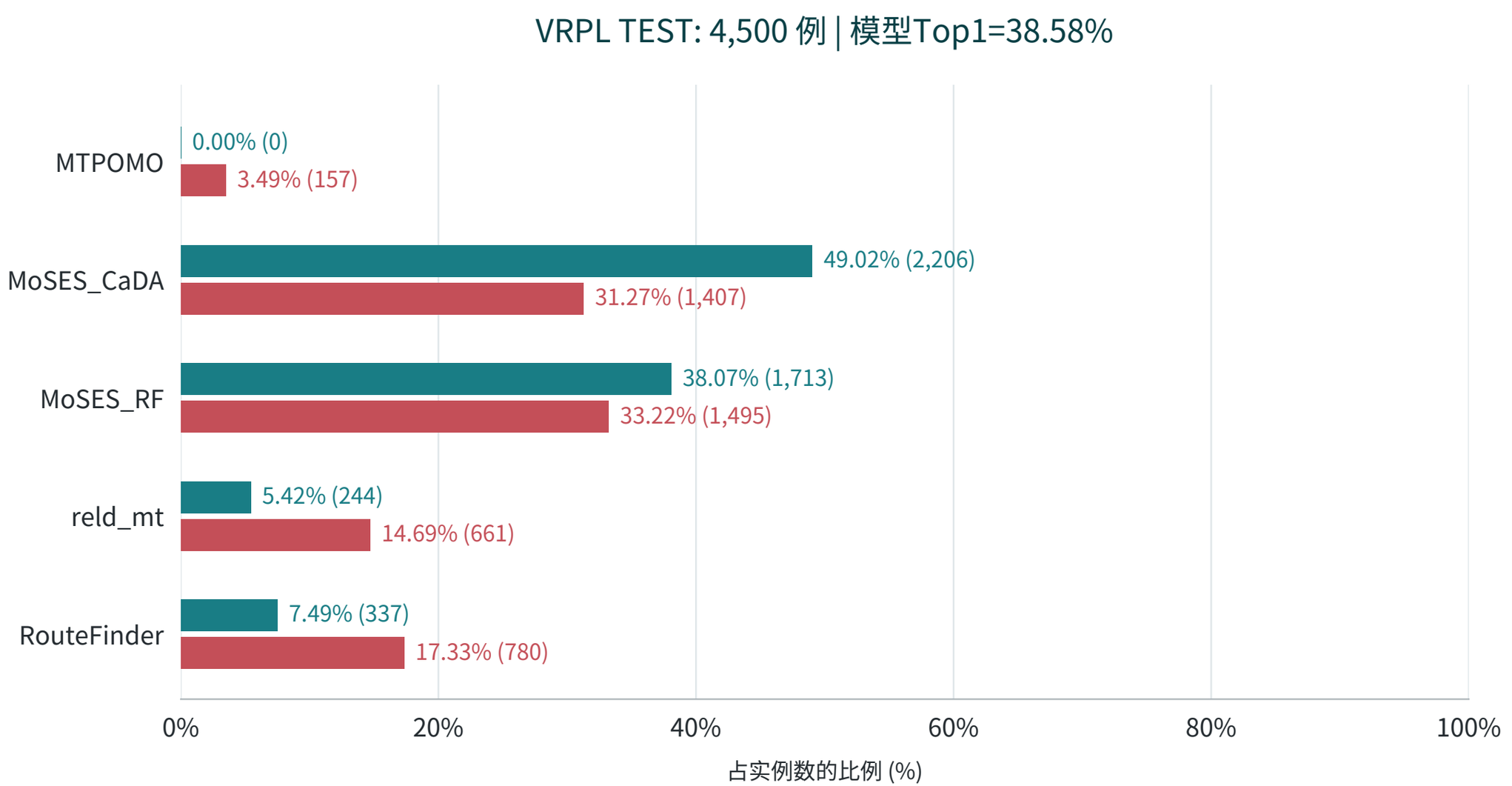
R127: only MVMOE deleted; BQ/LEHD and originally legal MoSES_RF retained; same VAL lock.

R127/model_only_seed2 | VRPL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 389dcc8b1817e6c9

新 TEST 数据版本: selection_data_new90k_no_mvmoe_seed2_20261112

标签: 原生 FP64 argmin after MVMOE deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

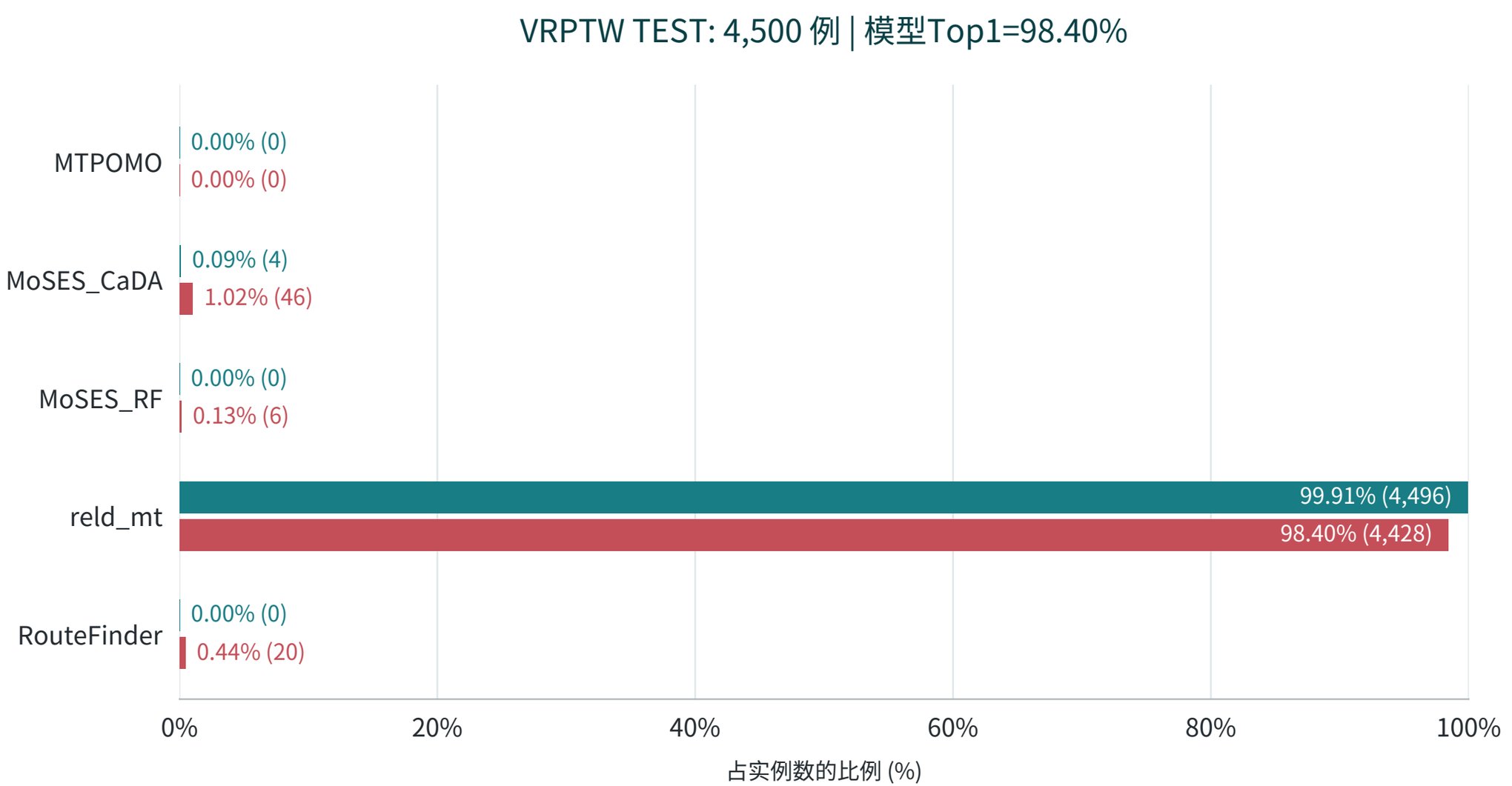
R127: only MVMOE deleted; BQ/LEHD and originally legal MoSES_RF retained; same VAL lock.

R127/model_only_seed2 | VRPTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 389dcc8b1817e6c9

新 TEST 数据版本: selection_data_new90k_no_mvmoe_seed2_20261112

标签: 原生 FP64 argmin after MVMOE deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

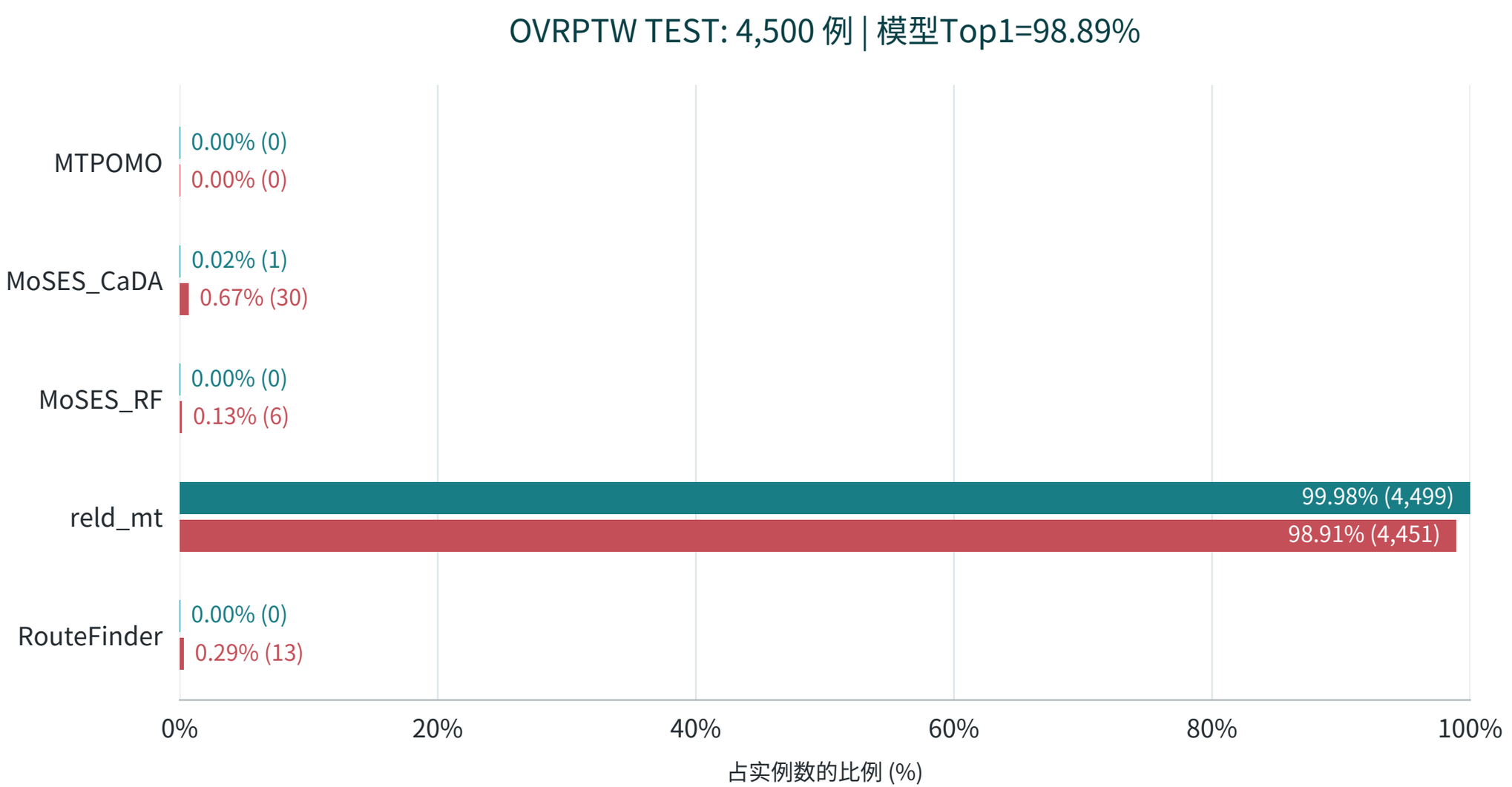
R127: only MVMOE deleted; BQ/LEHD and originally legal MoSES_RF retained; same VAL lock.

R127/model_only_seed2 | OVRPTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 389dcc8b1817e6c9

新 TEST 数据版本: selection_data_new90k_no_mvmoe_seed2_20261112

标签: 原生 FP64 argmin after MVMOE deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

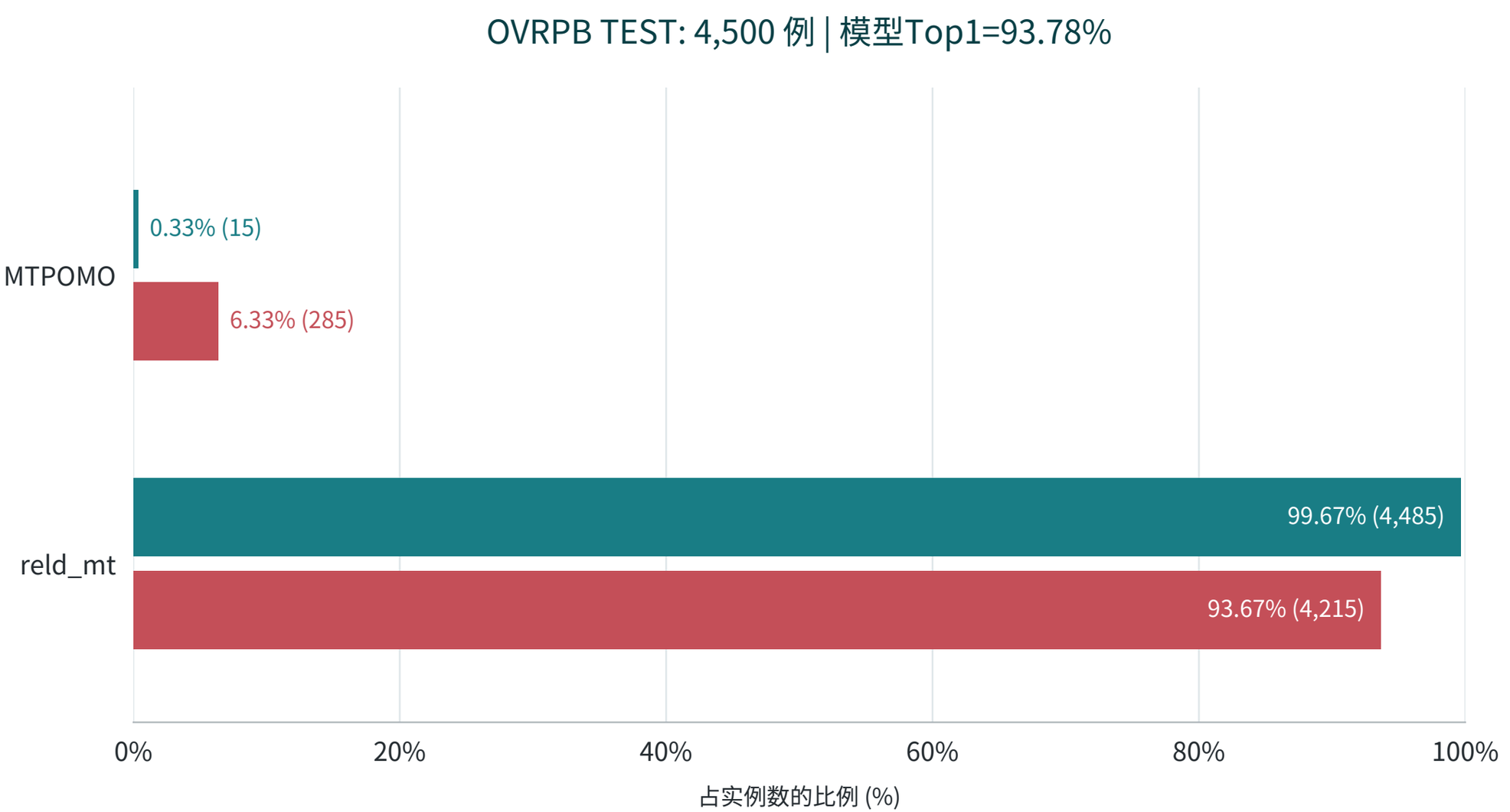
R127: only MVMOE deleted; BQ/LEHD and originally legal MoSES_RF retained; same VAL lock.

R127/model_only_seed2 | OVRPB | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 389dcc8b1817e6c9

新 TEST 数据版本: selection_data_new90k_no_mvmoe_seed2_20261112

标签: 原生 FP64 argmin after MVMOE deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

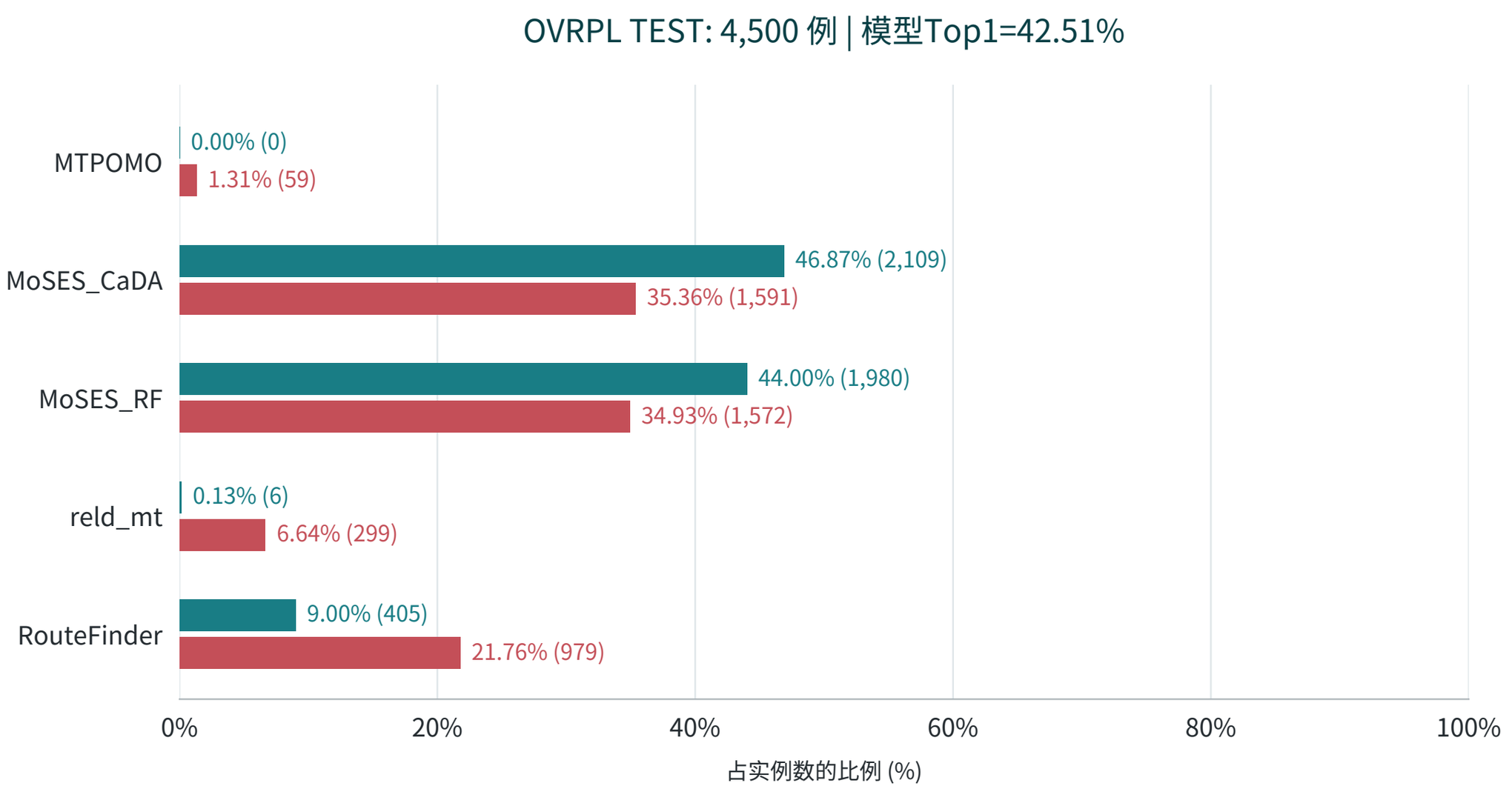
R127: only MVMOE deleted; BQ/LEHD and originally legal MoSES_RF retained; same VAL lock.

R127/model_only_seed2 | OVRPL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 389dcc8b1817e6c9

新 TEST 数据版本: selection_data_new90k_no_mvmoe_seed2_20261112

标签: 原生 FP64 argmin after MVMOE deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

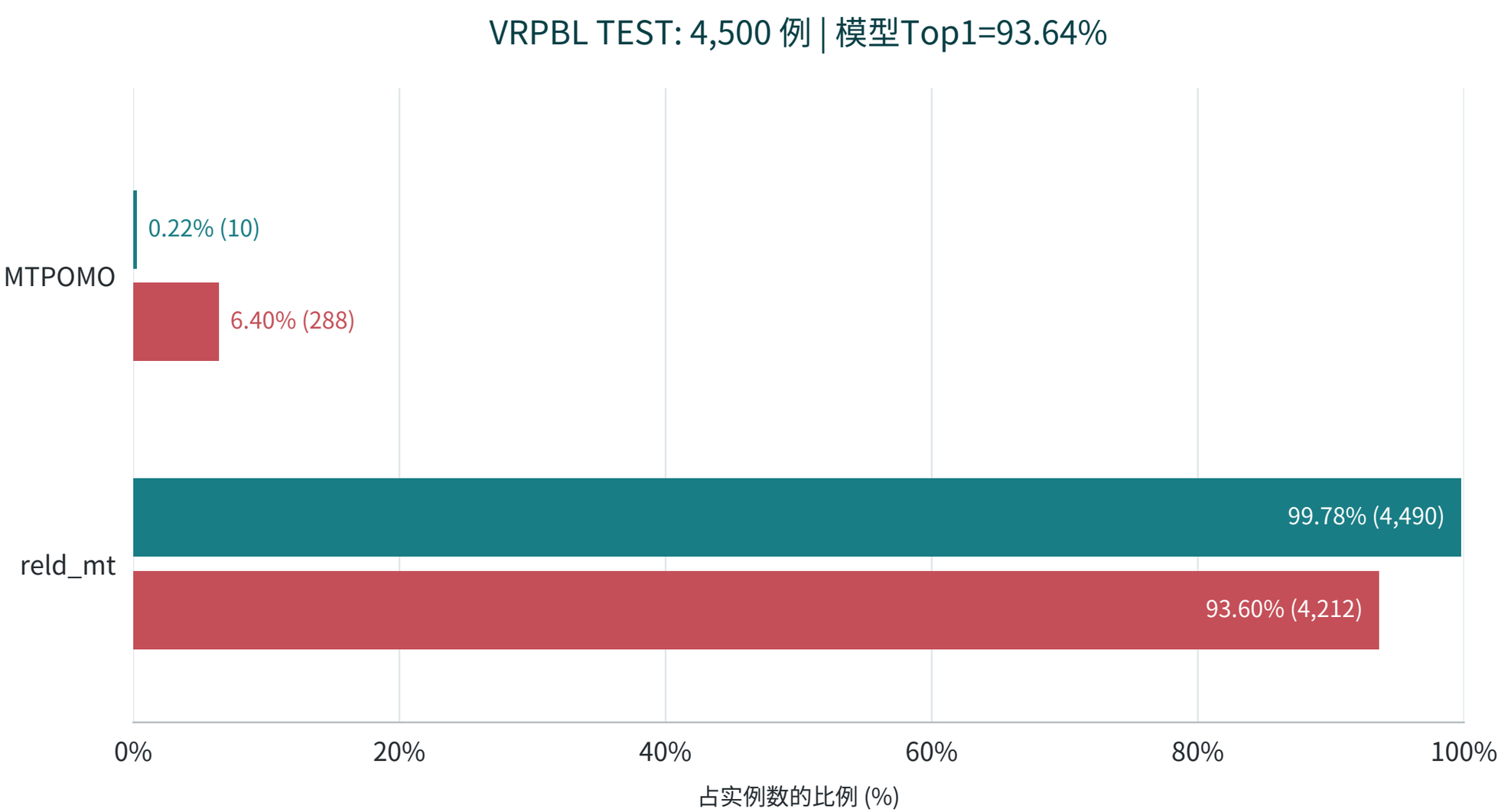
R127: only MVMOE deleted; BQ/LEHD and originally legal MoSES_RF retained; same VAL lock.

R127/model_only_seed2 | VRPBL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 389dcc8b1817e6c9

新 TEST 数据版本: selection_data_new90k_no_mvmoe_seed2_20261112

标签: 原生 FP64 argmin after MVMOE deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

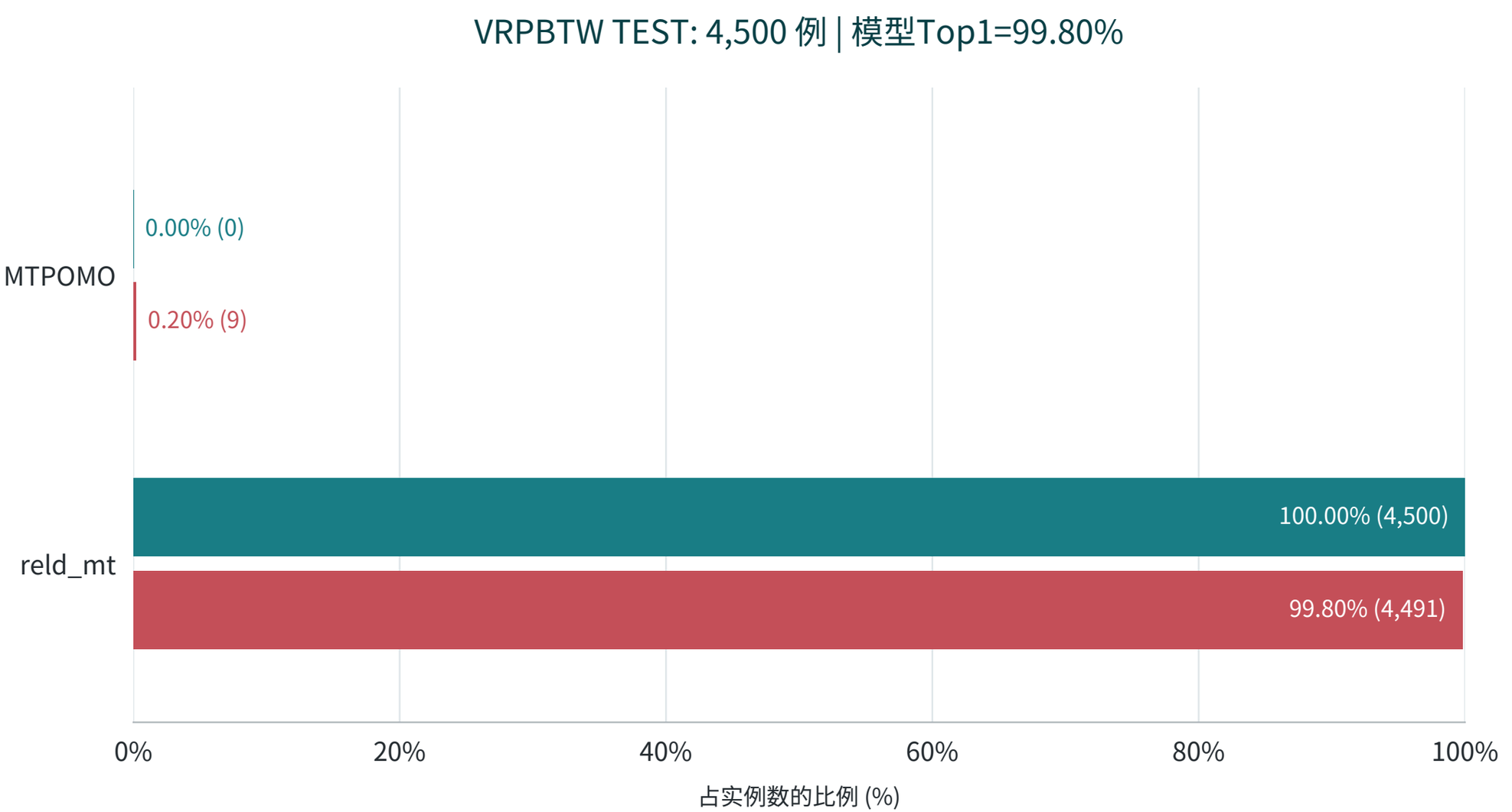
R127: only MVMOE deleted; BQ/LEHD and originally legal MoSES_RF retained; same VAL lock.

R127/model_only_seed2 | VRPBTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 389dcc8b1817e6c9

新 TEST 数据版本: selection_data_new90k_no_mvmoe_seed2_20261112

标签: 原生 FP64 argmin after MVMOE deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

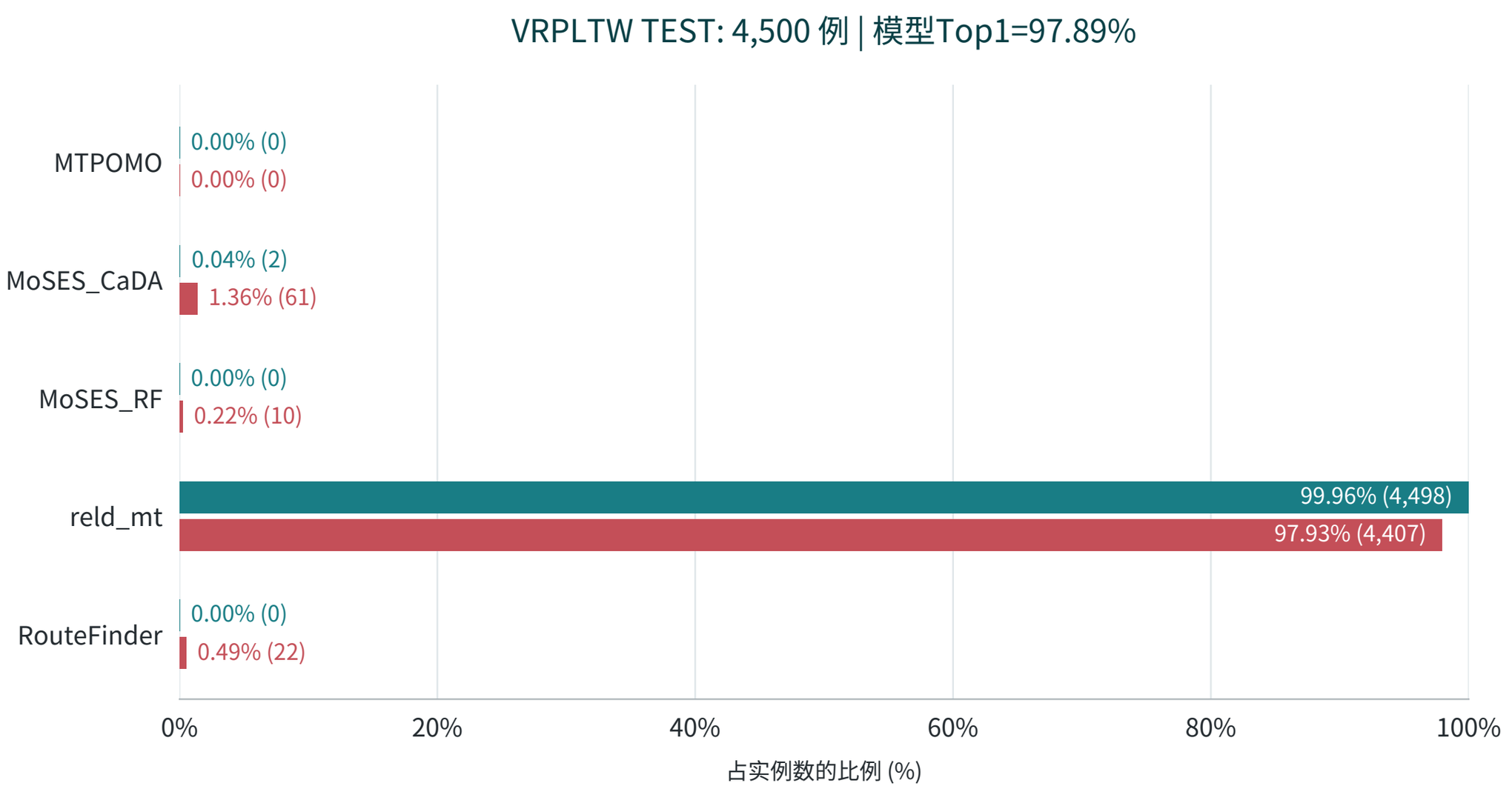
R127: only MVMOE deleted; BQ/LEHD and originally legal MoSES_RF retained; same VAL lock.

R127/model_only_seed2 | VRPLTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 389dcc8b1817e6c9

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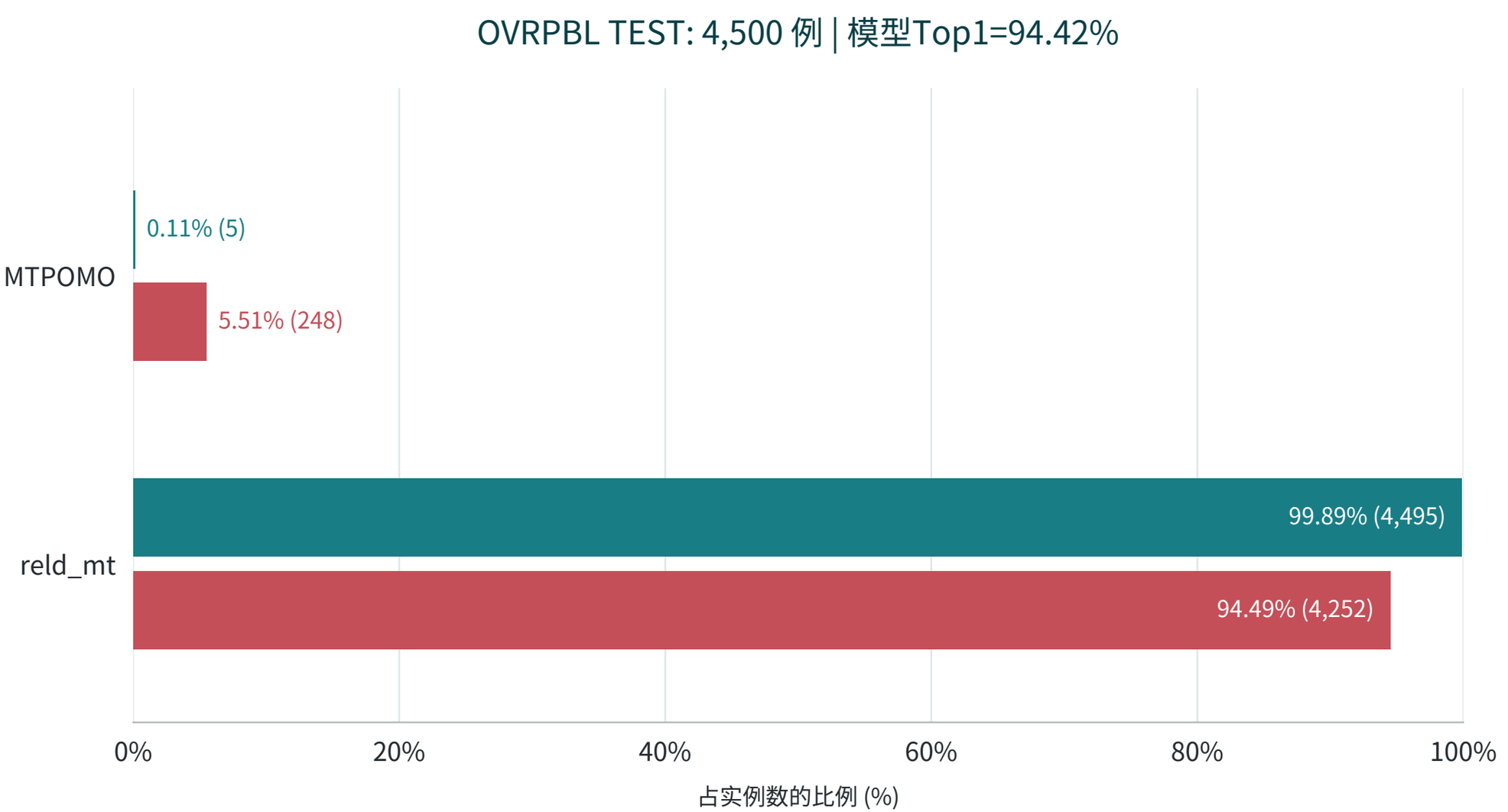
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R127/model_only_seed2 | OVRPBL | 方法分布

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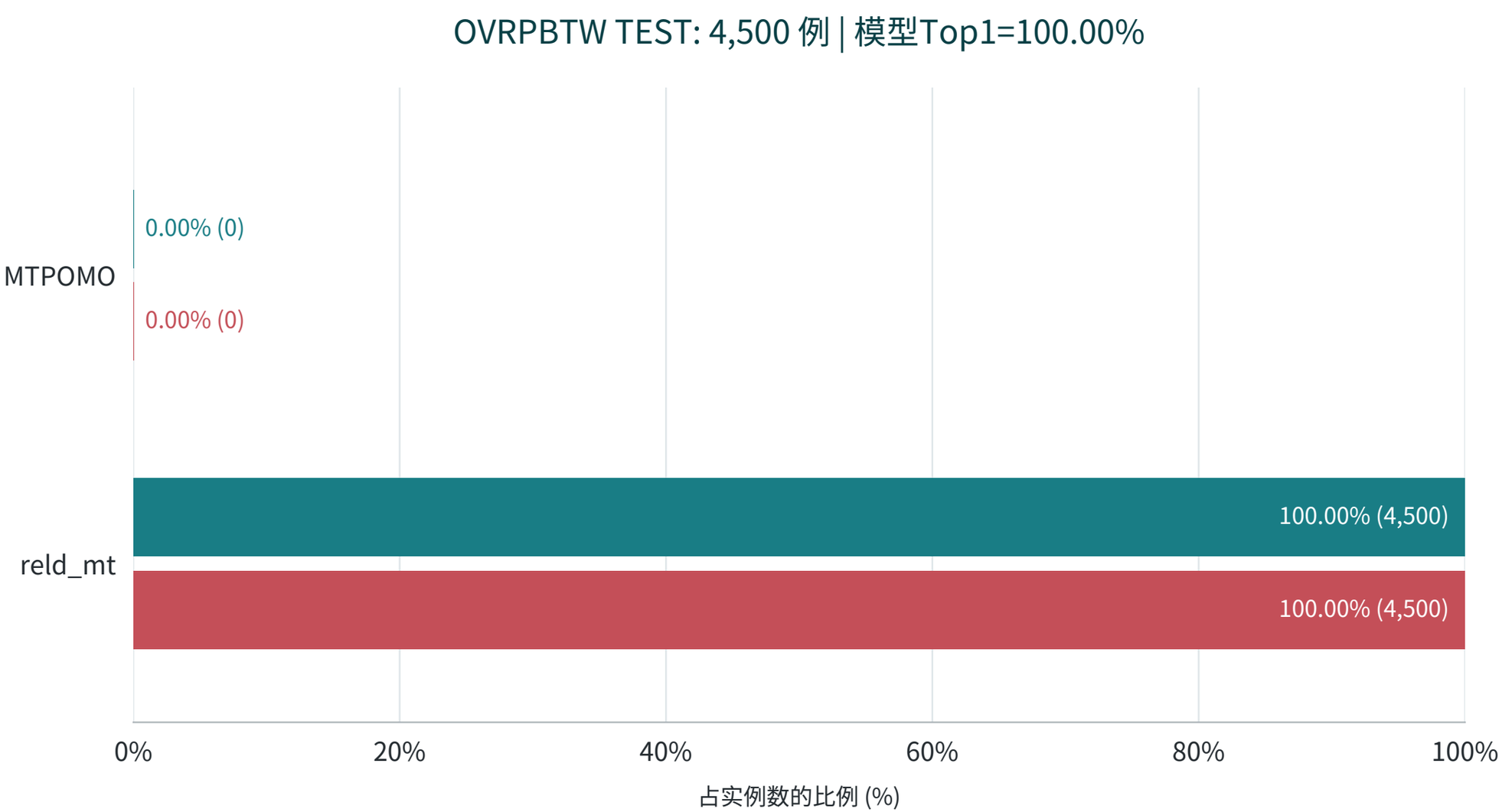
R127: only MVMOE deleted; BQ/LEHD and originally legal MoSES_RF retained; same VAL lock.

R127/model_only_seed2 | OVRPBTW | 方法分布

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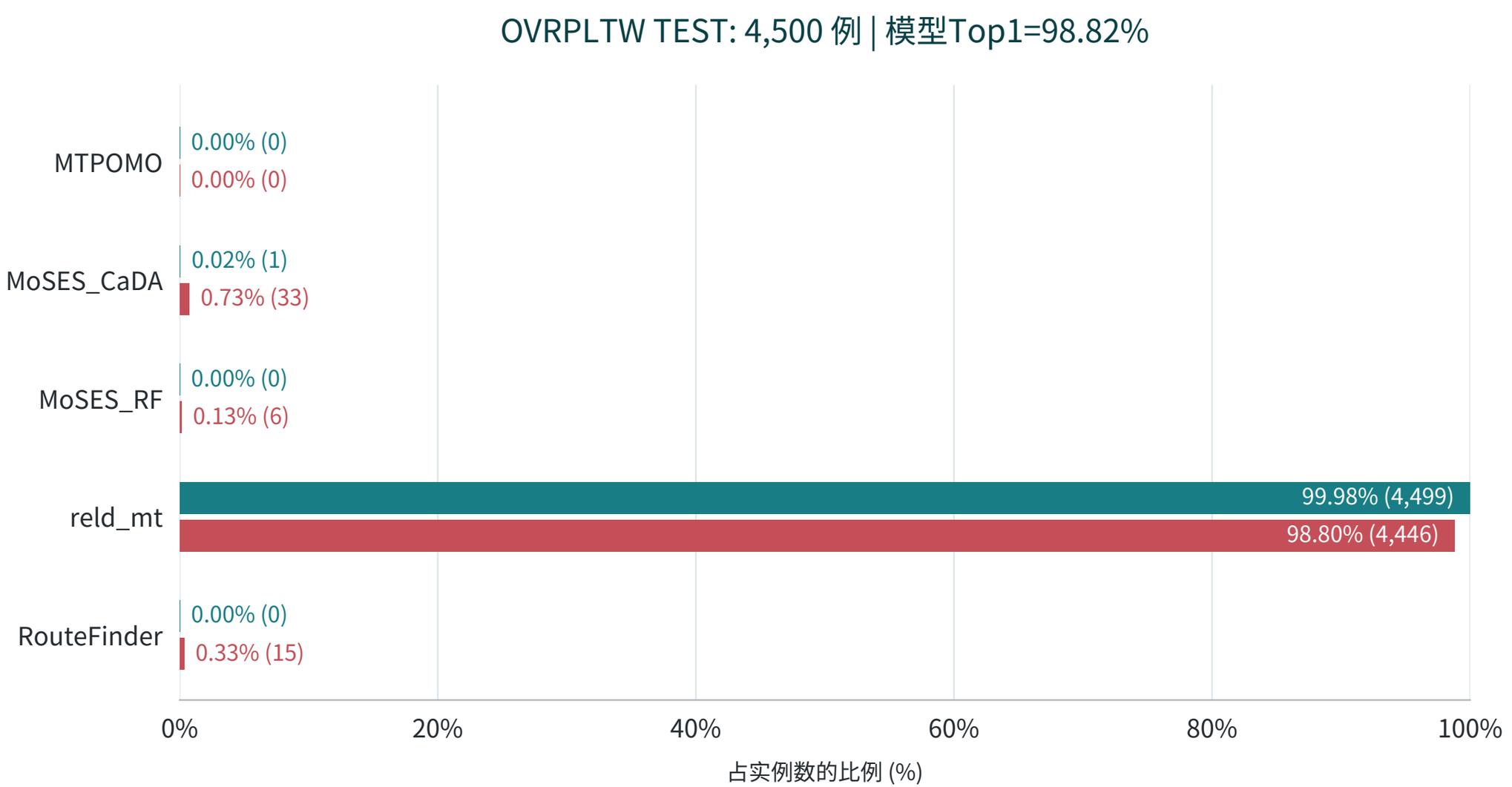
R127: only MVMOE deleted; BQ/LEHD and originally legal MoSES_RF retained; same VAL lock.

R127/model_only_seed2 | OVRPLTW | 方法分布

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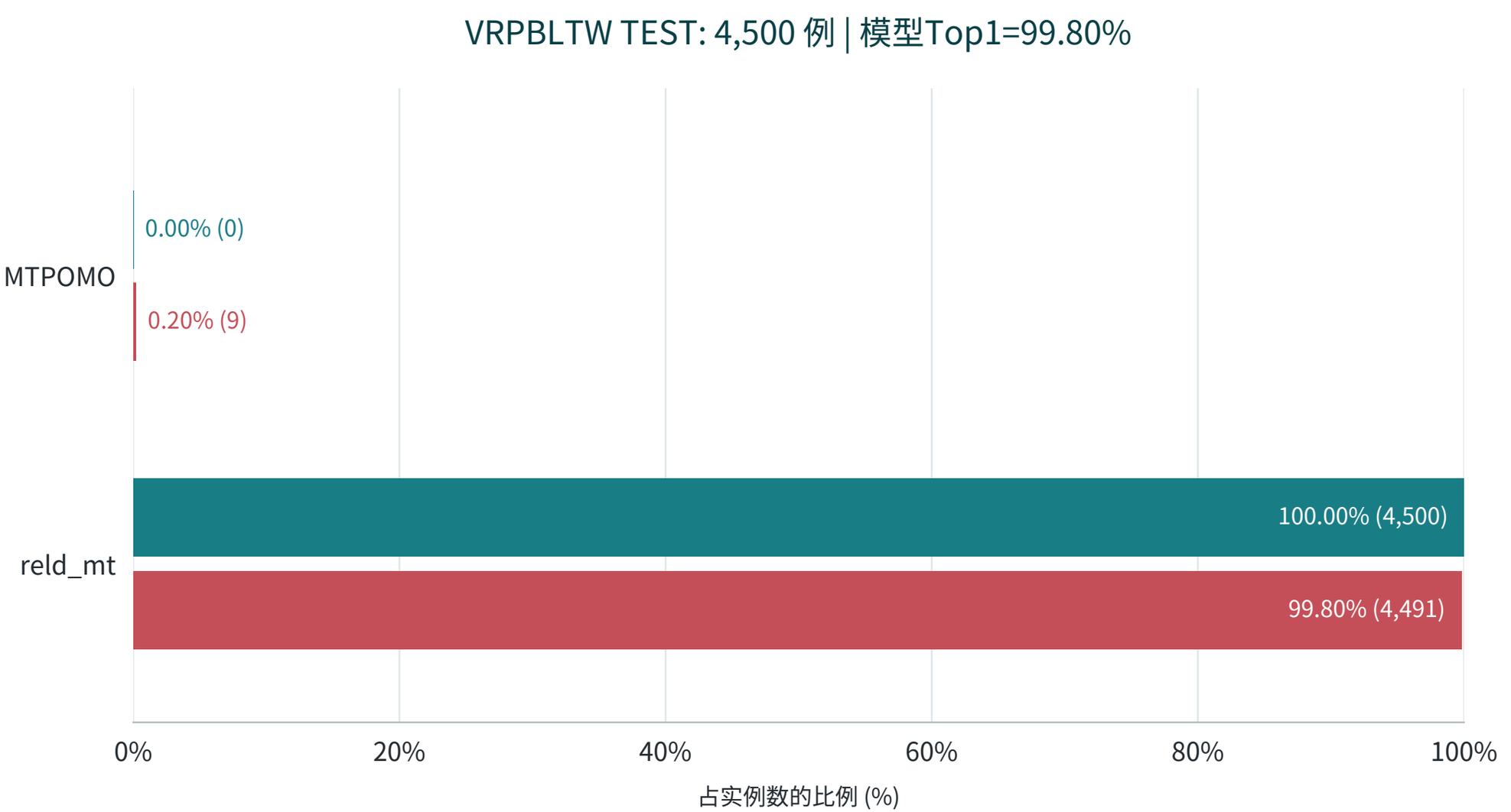
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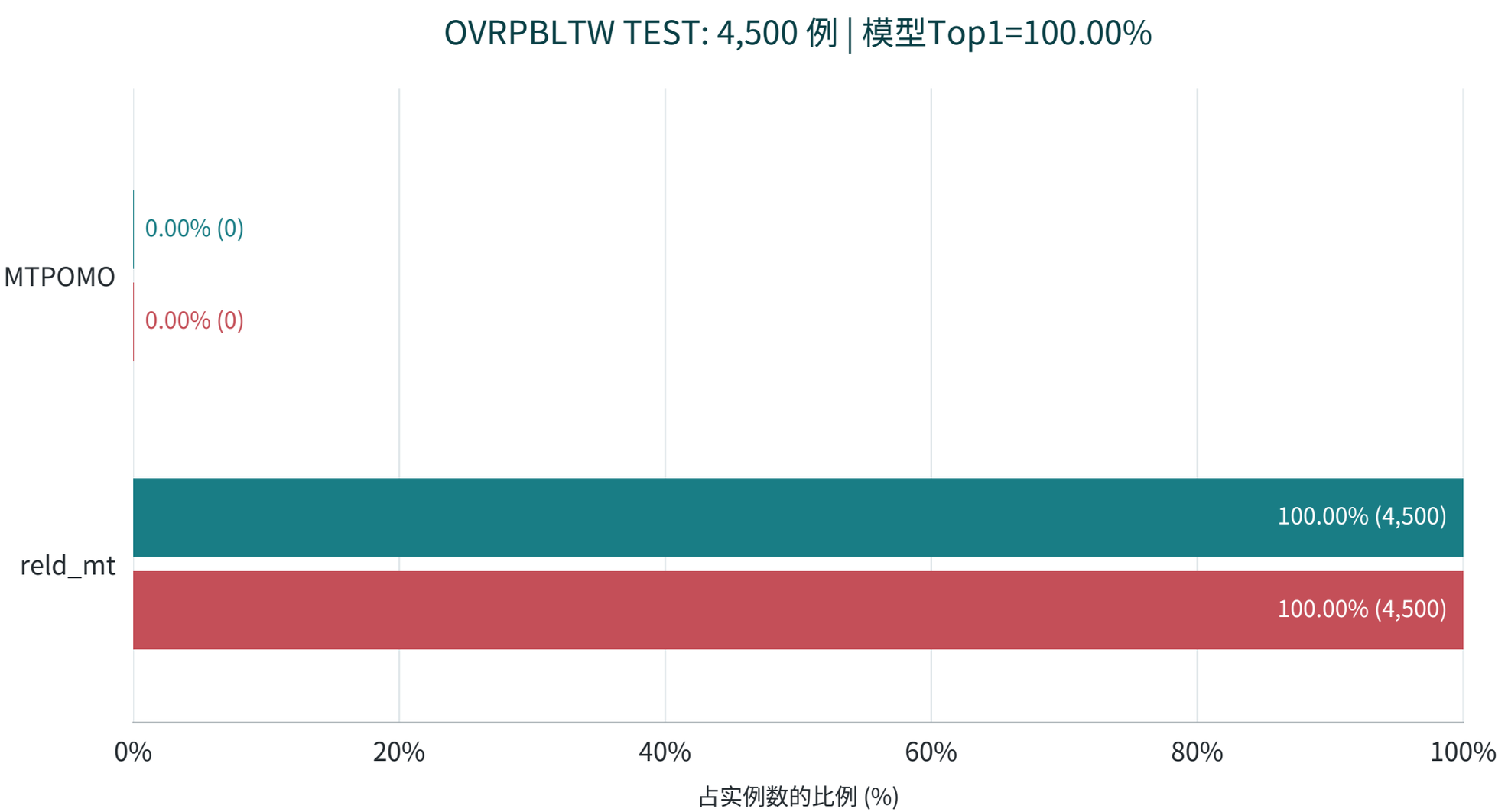
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